

Mannat Debnath

Agartala, Tripura, 799002 | 9233010261 | mannatdebnath05@gmail.com
[LinkedIn: mannat-debnath](#) | [GitHub: mannat-05](#)

EDUCATION

Sardar Vallabhbhai National Institute of Technology <i>Bachelor of Technology in Computer Science</i>	Surat, Gujarat <i>Aug 2024 – May 2028 (Expected)</i>
• CGPA: 8.58 / 10.0 • Relevant Coursework: Data Structures & Algorithms, OOP, Database Management Systems, Computer Networks, Information Security, Artificial Intelligence, Image Processing, Microprocessor Interfacing Techniques, Automata & Formal Languages.	

EXPERIENCE

Sardar Vallabhbhai National Institute of Technology <i>Undergraduate Research Assistant</i>	Surat, Gujarat <i>Dec 2025 – Present</i>
• Advisors: Prof. M.A. Zaveri and Dr. Anugrah Jain (Department of Computer Science) • Conducting research on Hyperspectral and Multispectral Image Super-Resolution to enhance visual reconstruction precision. • Implemented deep learning architectures (e.g., SRCNN) in PyTorch and benchmarked performance across a dataset of 10,000+ images. • Authoring two research papers (one conference paper and one journal article currently submitted/under review) detailing methodological advancements.	

PROJECTS

HALO (Heuristic Action-Locked Observation) <i>Python, OpenCV, Computer Vision</i>	July 2026
• 1st Place Winner in the Computer Vision Track at Hackathon. [GitHub Repository] • Engineered a real-time computer vision action tracking tool designed to monitor and lock specific tactical observations.	
Super Resolution using SRCNN <i>Python, OpenCV, PyTorch</i>	Dec 2025 – Jan 2026
• Built an SRCNN pipeline for single RGB image resolution enhancement, leveraging perceptual (VGG) loss. • Attained a PSNR of approx25 dB and SSIM of approx0.8, significantly suppressing random spatial noise. [GitHub Repository]	
Kala Sangam <i>React.js, Tailwind CSS</i>	Jun 2025 – Aug 2025
• Created a web platform mapping regional Indian art forms and showcasing the heritage of India and it's rich diversity. [Deployment Link]	

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL
Frameworks & Libraries: PyTorch, OpenCV, React.js, Tailwind CSS, Node.js
Developer Tools: Git, GitHub, VS Code, Antigravity, Kaggle, Postman
Core Concepts: Data Structures & Algorithms, DBMS, OOP, Artificial Intelligence, Computer Vision, Networking

ACHIEVEMENTS, CODING PROFILES & CERTIFICATIONS

Competitive Programming: LeetCode (500+ questions solved) [Profile] Codeforces (Specialist, Max Rating: 1417) [Profile]
Hackathon Winner: 1st Place (Computer Vision Track) for HALO project in a 48-hour hackathon by ACM RVCE. [Certificate]
CanSat Competition: Secured 3rd Place out of 20+ internal university teams.
Certifications: Image Processing using OpenCV and Python (Udemy). [Certificate]

EXTRACURRICULAR ACTIVITIES

NEXUS	Executive
• Mentored junior students on foundational Data Structures, Algorithms, and logic-building approaches.	
SCOSH	Executive
• Anchored an interactive research methodology workshop conducted by a Research head of Reliance hosted for over 50 undergraduate students.	